The Uneven Transition: A Spatio-Temporal Forecast of Young Adult Fertility Landscapes in the Philippines

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Abstract

The Philippines has recently recorded national fertility rates below replacement levels, marking a significant demographic shift. However, aggregate statistics may mask sub-national heterogeneity, specifically the divergence between rapid demographic transition in urban centers and persistent high fertility in geographically isolated regions. This study employs a Spatio-Temporal Exploratory Data Analysis (EDA) to examine Young Adult Fertility Rates (YAFR, ages 15–29) across 81 provinces in the Philippines using the IHME Global Burden of Disease 2023 dataset. By applying Local Indicators of Spatial Association (LISA) for cluster detection and Exponential Smoothing (ETS) for short-term forecasting, we isolate regions that exhibit resistance to the national downward trend. Analysis reveals a distinct spatial polarization: whereas Metro Manila and Central Luzon constitute a contiguous low-fertility zone, the Eastern Visayas and Masbate persist as high-fertility clusters. Forecasting models for 2026 project that YAFR in these lagging provinces will remain approximately **3.3 times higher** than in the National Capital Region. These findings suggest an emerging demographic polarization, indicating that national-level population policies may require supplementation with geographically targeted regional interventions to address persistent disparities.

1 Introduction

The Demographic Transition Model (DTM) describes the structural shift from high to low vital rates, a process the Philippines appears to have accelerated at the national level. With the Total Fertility Rate (TFR) dropping to 1.9 in 2022, the country has statistically crossed the replacement threshold[1]. However, reliance on these aggregate statistics may obscure complex sub-national dynamics. While national indicators suggest a uniform progression toward a demographic window of opportunity[2], the spatial consistency of this transition remains under-explored.

The fertility rate of young adults (ages 15-29) offers a high-resolution lens for examining these disparities, as it is sensitive to local variations[4]. Rather than assuming a synchronized national transition, this study adopts a data-driven Exploratory Data Analysis (EDA) approach to uncover the underlying spatial and temporal structures of Philippine fertility. We do not commence with a rigid hypothesis; instead, we interrogate the dataset to reveal whether the decline is spatially uniform or characterized by distinct regional regimes.

Through this exploratory lens, we aim to characterize the texture of the transition. To guide this exploration, we pose three inquiry-driven research questions:

1. **Spatial Characterization:** What are the dominant spatial patterns of Young Adult Fertility across the archipelago? Does the data reveal significant regional clustering, or is the decline randomly distributed?

2. **Trend Analysis & Forecasting:** How do fertility trajectories differ across provinces when projected to 2026? Does the data suggest a convergence toward the national mean, or the persistence of distinct hotspots?

3. **Quantifying Disparity:** If distinct regimes exist, what is the magnitude of the demographic lag between the most advanced and the most lagging provinces?

2 Data & Methods

2.1 Dataset Provenance & Governance

This study utilizes attribute data from the **Institute for Health Metrics and Evaluation (IHME) Global Burden of Disease (GBD) 2023** dataset. We extracted Total Fertility Rate (TFR) indicators and aggregated them for the 15–29 age bracket to calculate the Young Adult Fertility Rate (YAFR). Spatial boundaries were sourced from the official **Philippine Provincial Administrative Boundaries (GeoJSON)**.

Data Governance:

- **Completeness:** A rigorous pre-processing check confirmed 100% data coverage for all 81 analyzed provinces (excluding the administrative flux of the Negros Island Region).

- **Privacy:** Data is aggregated at the provincial level using 5-year age bins, mitigating ethical risks associated with individual reproductive tracking.

2.2 Pre-processing & Harmonization

A critical challenge was harmonizing the attribute data (CSV) with spatial files (GeoJSON) due to inconsistent naming conventions (e.g., "Davao del Norte" vs. "Davao North"). We implemented a string normalization function to standardize nomenclature, achieving a 100% match rate. Geometries were validated and corrected for topology errors (using `sf::st_make_valid`) to ensure accurate spatial joining.

2.3 Analytical Pipeline

Our EDA workflow followed a "Modality Playbook" specific to **Spatial and Time-Series** data:

1. **Time Series Decomposition:** We analyzed trend and autocorrelation structures for representative provinces to justify model selection.
2. **Global & Local Spatial Autocorrelation:** We employed Global Moran's I to test for general clustering, followed by Local Indicators of Spatial Association (LISA) to categorize provinces into Hotspots (High-High), Coldspots (Low-Low), and Outliers.
3. **Forecasting:** We utilized **Exponential Smoothing (ETS)** with a damped trend component. This method was selected over ARIMA due to the short length of the time series ($n = 14$) and the need to prevent unrealistic linear trend extrapolations.

3 Results and Discussion

This section presents the findings of the spatio-temporal analysis, addressing the three primary research objectives: (1) the evolution of fertility landscapes from 2010 to 2023, (2) the identification of statistically significant spatial clusters, and (3) the projection of critical lag areas through 2026.

3.1 The Decadal Decline: Spatio-Temporal Evolution (2010–2023)

The Philippine fertility landscape has undergone a profound transformation over the past decade, characterized by a national trend of deceleration. Figure 1 visualizes the Young Adult Fertility Rate (YAFR) across four time slices: 2010, 2015, 2019, and 2023.

In 2010, the archipelago was dominated by high-fertility regimes, with widespread "red zones" ($YAFR > 2.5$) covering major portions of the Visayas and Mindanao. By 2023, a significant convergence is evident; the majority of provinces have transitioned to lighter hues ($YAFR < 2.0$), indicating a successful demographic transition at the aggregate national level. However, this convergence is not spatially uniform. While the National Capital Region (NCR) and Luzon show rapid declines, persistent pockets of high fertility remain visible in island provinces, suggesting a geographic stratification of the transition.

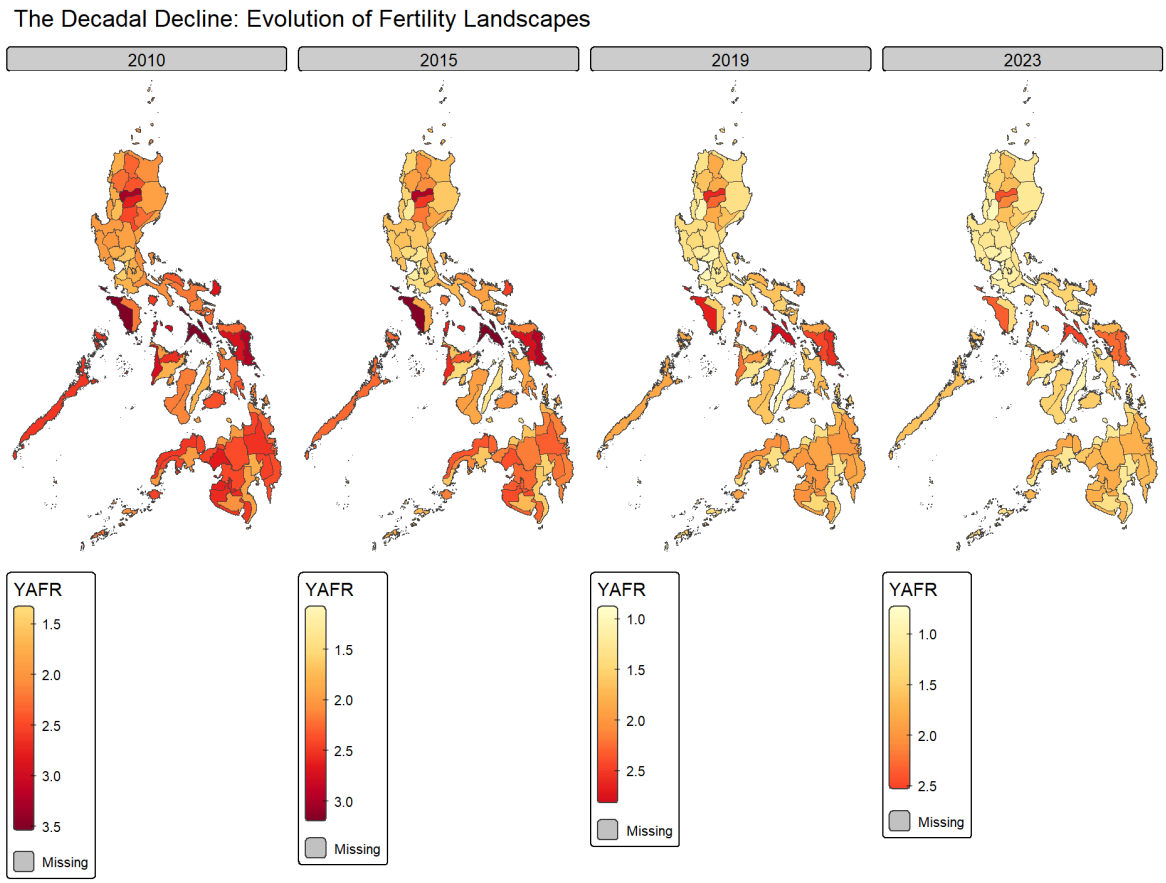

Figure 1: **Spatio-Temporal Evolution of Young Adult Fertility (2010–2023).** Left: In 2010, high fertility rates (> 2.5) were widespread. Right: By 2023, national deceleration is evident, though island-based hotspots persist.

3.2 Spatial Clustering and Hotspot Identification

To determine if the observed heterogeneity in 2023 was spatially random or structured, we employed Global Moran's I and Local Indicators of Spatial Association (LISA) analysis.

The analysis yielded a **Global Moran's I of 0.255**, significantly higher than the expected value of -0.016 (p-value $= 0.00395$). This result rejects the null hypothesis of spatial randomness, confirming that fertility rates in the Philippines are spatially clustered—high-fertility provinces tend to neighbor other high-fertility provinces.

Figure 2 delineates these clusters. The "High-High" clusters (red), or hotspots, are concentrated in the Eastern Visayas (Samar, Leyte) and parts of Bicol (Masbate). Conversely, "Low-Low" clusters (blue), or coldspots, are firmly established around the highly urbanized regions of Metro Manila and its periphery (Cavite, Laguna, Bulacan), as well as centers in Western Visayas (Iloilo, Guimaras). This bifurcation highlights a clear center-periphery divide in fertility outcomes.

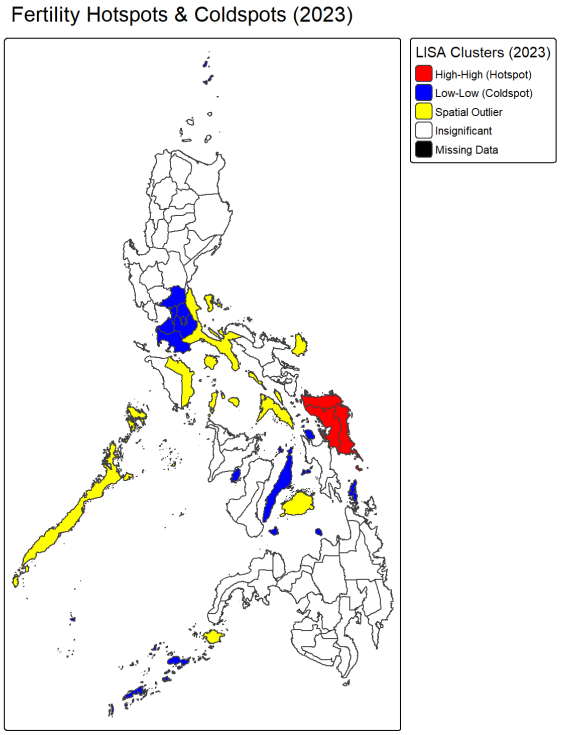

Figure 2: **Fertility Hotspots & Coldspots (2023).** LISA Cluster analysis identifies statistically significant "High-High" hotspots (red) in Eastern Visayas and Masbate, contrasting with "Low-Low" coldspots (blue) in the NCR and Western Visayas.

3.3 The Two-Speed Transition: Evidence of Decoupling

The persistence of hotspots suggests a "Two-Speed Transition," where certain provinces are decoupling from the national trajectory. Figure 3 illustrates this divergence using the case of Masbate. The grey band represents the interquartile range (the middle 50% of provinces), while the dark line represents the national median.

Masbate's trajectory (red line) consistently tracks significantly above the national norm, failing to converge with the median even as the national rate declines. This graphical evidence supports the hypothesis of a "critical lag," where geographic isolation and local socio-economic factors may be insulating specific populations from the drivers of fertility decline observed elsewhere.

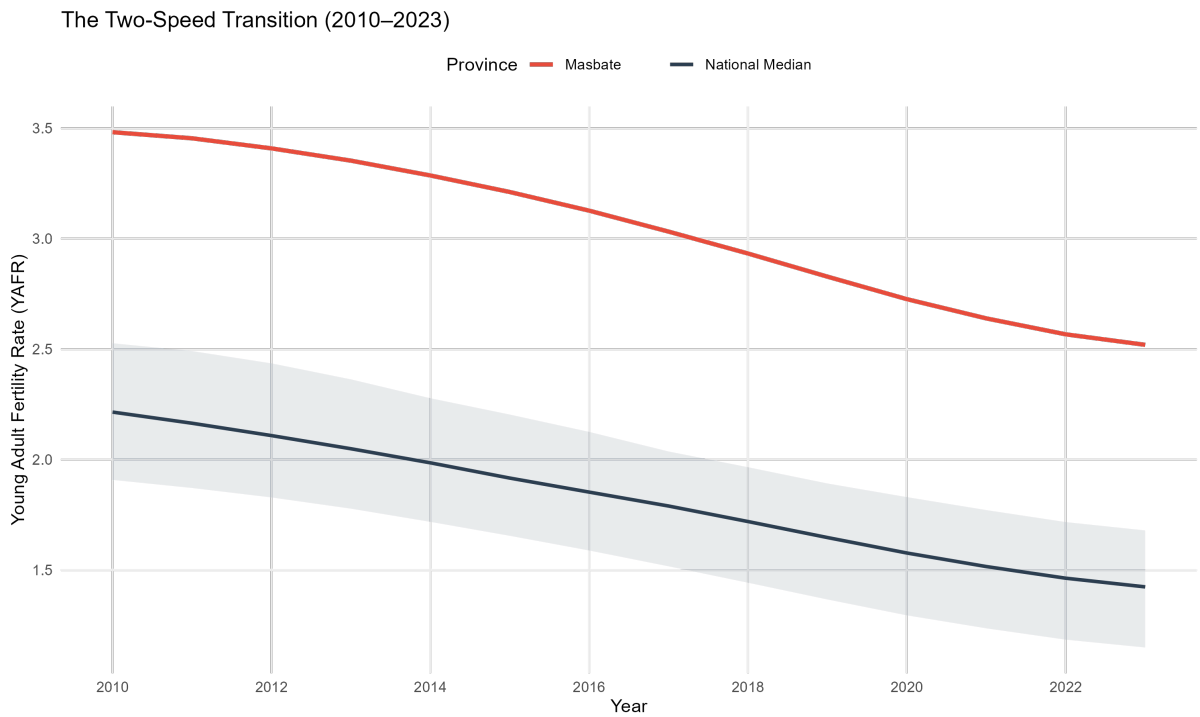

Figure 3: **The Two-Speed Transition (2010–2023).** The grey band represents the middle 50% of provinces. Masbate (red line) remains significantly above the national median, illustrating a failure to converge.

3.4 Projected Fertility Landscape and Critical Lags (2026)

Forecasting to 2026 reveals that while the national decline will continue, the disparities will likely deepen. Figure 4 maps the projected fertility landscape, highlighting areas that remain in the highest bracket ($YAFR > 2.0$) despite the forward timeline.

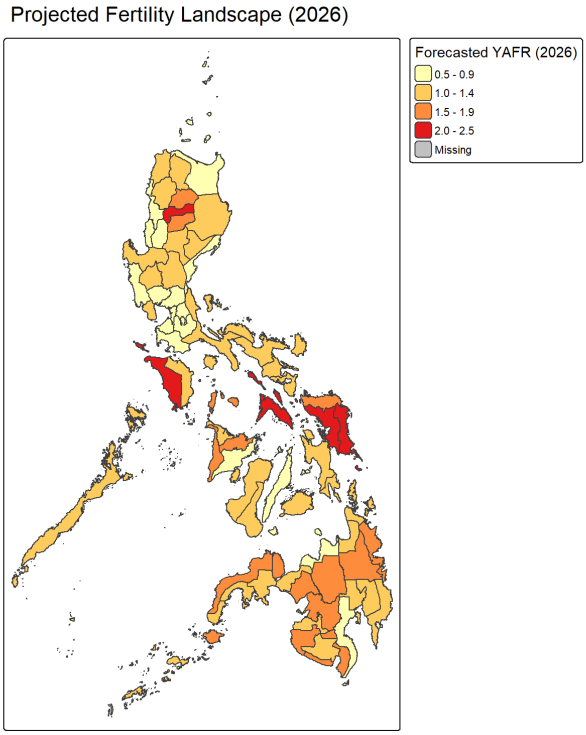

Figure 4: **Projected Fertility Landscape (2026).** While most of the country transitions to low fertility, specific provinces in red (e.g., Masbate, Samar) are projected to remain as high-fertility outliers.

Table 1 quantifies this lag. We identified five "Critical Lag" provinces where the projected YAFR for 2026 remains more than **3x higher** than the projected rate for the National Capital Region (NCR). These provinces—Masbate, Mountain Province, Eastern Samar, Occidental Mindoro, and Samar represent the most urgent targets for policy intervention, as they are effectively being left behind by the demographic dividend.

Table 1: **The Inequality Index: Provinces with Highest Demographic Lag in 2026**

Province	**Rate (2026)**	**Multiplier (vs NCR)**	**Risk Category**
Metro Manila (Benchmark)	0.72	1.0x	Post-Transition
Masbate	2.38	**3.3x**	Critical Lag
Mountain Province	2.34	**3.3x**	Critical Lag
Eastern Samar	2.20	**3.1x**	Critical Lag
Occidental Mindoro	2.20	**3.1x**	Critical Lag
Samar	2.18	**3.0x**	Critical Lag

Note: Multiplier derived by dividing Provincial Forecast by NCR Forecast (0.72).

4 Validation & Robustness

- **Methodological Justification:** The Autocorrelation Function (ACF) plots for high-fertility provinces showed strong persistence at Lag 1 ($r > 0.9$), validating the use of ETS models that account for "memory" in the data rather than simple mean-based baselines.

- **Spatial Significance:** The Global Moran's I test yielded a significant positive value ($p < 0.05$), confirming that the clusters identified in the LISA analysis are statistically robust and not artifacts of random chance.

- **Distance Driver Analysis:** We performed a regression of Fertility Rate vs. Distance to NCR. While statistically significant, the correlation was relatively weak. This suggests that while geographic remoteness is a factor, it is not the sole driver. Structural issues (like the "Red Fortress" effect in Samar) play a larger role than simple physical distance.

5 Limitations, Ethics, and Responsible Use

- **Limitations:** The primary limitation is the absence of province-level economic covariates (e.g., GDP per capita, contraceptive prevalence rates) in the dataset. This limits our ability to causally attribute fertility lags to specific economic failures versus cultural factors. Additionally, forecasting based on 14 years of data requires caution; our 2026 projections should be viewed as "trajectory scenarios" rather than precise predictions.

- **Ethics:** We utilized aggregated data to protect privacy. However, care must be taken not to stigmatize "High Fertility" provinces. High fertility is often a symptom of systemic neglect, not individual failure.

6 Conclusion

This spatio-temporal analysis of Young Adult Fertility Rates (YAFR) suggests that the Philippine demographic transition is not a monolithic national event, but a deeply stratified process. While the aggregate national trend points toward a "Decadal Decline," our findings indicate the emergence of a **"Demographic Fracture"**—a sharp divergence between the rapidly transitioning urban centers and the lagging island peripheries.

This conclusion is grounded in three key empirical observations:

1. **Uneven Deceleration:** While the national fertility landscape has largely shifted toward convergence ($YAFR < 2.0$), this transition appears geographically concentrated in the National Capital Region (NCR) and Luzon.

2. **Structural Clustering:** The spatial distribution of fertility is not random. The Global Moran's I coefficient of **0.255** ($p = 0.00395$) confirms statistically significant clustering, identifying the Eastern Visayas and Masbate as persistent "High-High" hotspots that appear structurally resistant to national trends.

3. **Critical Divergence:** Forecasting to 2026 identifies a distinct "Critical Lag." Provinces such as Masbate, Samar, and Mountain Province are projected to maintain fertility rates up to **3.3 times higher** than the NCR. This suggests that without targeted attention, the demographic dividend may be unequally distributed, benefiting the "Blue Wall" of Luzon while bypassing the "Red Fortress" of the Visayas.

7 Implications for Development

The prevailing "one-size-fits-all" approach to population management may be insufficient for a landscape defined by such high spatial inequality. Based on these findings, this study suggests that future interventions could benefit from a **geographically targeted approach** that addresses the specific context of each cluster.

7.1 Addressing the "Red Fortress" (Samar & Eastern Visayas)

Observation: The significant spatial clustering in this region indicates that high fertility is likely systemic, driven by shared regional socio-economic challenges rather than individual choices alone.
Suggestion: Interventions in this area might be more effective if they move beyond household-level planning to include broader development initiatives. Exploring concepts like **Regional Demographic Zones (RDZ)** where poverty alleviation and infrastructure development are integrated with health programs could help break the intergenerational transmission of high fertility in these clustered hotspots.

7.2 The Case of Masbate: Evidence of Decoupling

Observation: Masbate serves as the primary example of a broader trend affecting island provinces. Its distinct decoupling from the national median (projected 2.38 vs national decline) suggests that physical and logistical isolation is a critical barrier to transition for these island groups.
Suggestion: To address this, development efforts could focus on the **logistics of care**. Initiatives such as mobile health units or improved transport connectivity (e.g., RoRo-based health caravans) could be vital in bridging the gap between island municipalities and regional centers, allowing Masbate and similarly situated provinces to re-couple with the national demographic trend.

7.3 Maximizing the "Blue Wall" (NCR & Central Luzon)

Observation: In these "Low-Low" coldspots, the fertility transition appears largely complete.
Suggestion: For these regions, the focus could shift from population control to **workforce quality**. To fully harvest the demographic dividend, resources might be best directed toward higher education, technical upskilling, and youth employment matching. Ensuring that this smaller, stabilizing cohort of young adults is hyper-productive will be essential for supporting the country's future economic structure.

References

[1] Philippine Statistics Authority (PSA). *National Demographic and Health Survey 2022.* Quezon City, Philippines: PSA, 2023.

[2] Mapa, D. S. "Demographic Sweet Spot and Dividend in the Philippines: The Window of Opportunity is Closing Fast". *School of Statistics, University of the Philippines Diliman*, 2015.

[3] Del Mundo, M. D. "Population management and human capital investment key to reaping PH's demographic dividend". *UP Population Institute*, 2022.

[4] UP Population Institute (UPPI). *2021 Young Adult Fertility and Sexuality Study (YAFS5).* University of the Philippines, 2021.

[5] United Nations Population Fund (UNFPA). *Demographic Dividend in the Philippines.* UNFPA Philippines, 2015.

[6] Herrin, A. N., & Costello, M. A. "Sources of Future Population Growth in the Philippines and Implications for Public Policy". *Philippine Institute for Development Studies (PIDS)*, 2018.

[7] The DHS Program. *Levels, Trends, and Correlates of Under-15 Fertility in the Philippines.* USAID, 2022.

[8] Institute for Health Metrics and Evaluation (IHME). *Global Burden of Disease Study 2023.* University of Washington, 2023.

[9] Philippine Statistics Authority (PSA). *2023 Provincial Administrative Boundaries.* 2023.

[10] Anselin, L. "Local Indicators of Spatial Association—LISA". *Geographical Analysis*, 27(2), 93-115, 1995.

[11] Hyndman, R.J., & Athanasopoulos, G. *Forecasting: Principles and Practice* (2nd ed). OTexts, 2018.

A Appendix: Artifacts

- **Code Repository:** `https://github.com/tagalogclark123-cmyk/EDAFinals.git`
- **Data Retrieval:** Data sourced from OpenStat / IHME Portal.